

Car Competition

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Student Id

Introduction

For the final car competition, you will be extending your work from the midterm mind-controlled car demo by controlling the car to turn right and left using hybrid-BCI approach. Hybrid-BCI means you can use information or features other than EEG (e.g., artifacts, additional measurements on EEG devices) as the control signal. The competition objective is to control your car and finish a track set up by the TAs.

Schedule

- EEG device test and algorithm development (11/21 ~ 12/11)
 - 12:00~13:00 every Friday (after class)
 - Scheduled time with TA
 - Due to limited space in the lab 616, please use the form to make an [reservation](#) if you wish to come to the lab to test equipment.
- Final competition (1 week)
 - 12/12 , adjustments on site is allowed
- Technical Report
 - 12/19

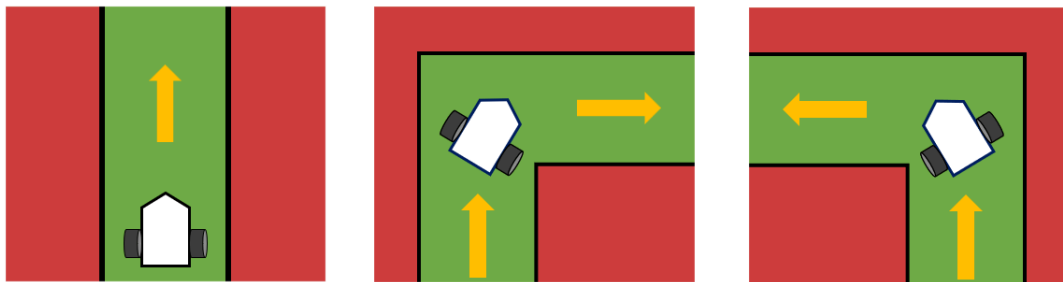
Rules

- Controls allowed for Hybrid BCI *
 - EEG**
 - Physiological artifacts (e.g., EOG, EMG, ECG)
 - Gyroscope of EEG device (Acceleration and angular acceleration on X, Y, Z-axis)
- Each group will have 3 trial opportunities in the competition, **each trial less or equal to 10 minutes (not including the penalty).**
- Out of bound penalty: **+1 min** each time
- Competition ends at 13:00.

* Keyboard control, physical displacement or any other controls not listed above are all forbidden
**Since this is BCI course, EEG is required.

Grading Policy

- Basic functionality (0%)
 - Show the sources / algorithms / logics your group utilized for the controls
 - Move forward
 - Turn left
 - Turn right



- Tiers (80%)
 - Basic tier scoring + time cost ranking
 - **A**: 95 points + in tier ranking by 3 points interval
 - **B**: 75 points + in tier ranking by 3 points interval
 - **C**: 60 points + in tier ranking by 2 points interval
 - **D**: 50 points + in tier ranking by 2 points interval

	EEG BCI	Hybrid BCI
Checkpoint 1	C	D
Checkpoint 2	B	C
Checkpoint 3	A	B

(For example, if 3 teams reach tier B, the grades will be $75+3*2=81$, $75+3=78$, 75 respectively base on their implementation types (EEG > Hybrid) and finish time)

- Technical Report (20%)

Technical Report

Please submit a report by group answering the following:

- Method: Your strategies implementing the BCI car, including EEG/other signal feature, preprocessing, classifier, control method etc.
- Discussion 1: The challenges and difficulties you encountered during the process and how you solve them.
- Discussion 2: Other things you wish to discuss, such as contribution.
- Contribution chat: Contributions of each group member.

Only one team member per group will have to submit the report to the 'Lab3: Car Competition' assignment on E3. Please submit a zip file containing (1) the report pdf file and (2) the code for the BCI car. Please name your zip file as lab3_GroupX.zip

	Gr1	Gr2	Gr3	Gr4	Gr5
Trial 1	Hybrid C1 5 mins (12 penalty)	Hybrid	EEG 5 mins (2 mins penalty)		Hybrid C2 5 mins (8 mins penalty)
Trial 2	* Hybrid 2:52 (9 mins penalty)	Hybrid C1 5 mins (4 mins penalty)	* EEG 5 mins (1 mins penalty)	* EEG 5 mins (2 mins penalty)	* Hybrid 5 mins (4 mins penalty)
Trial 3	Hybrid	* Hybrid 5:58 (6 mins penalty)	EEG C1 5 mins (1 mins penalty)	EEG C1 5 mins (3 mins penalty)	Hybrid C2 5 mins (9 mins penalty)
	75+3=78	75+6=81	75+3=78	60+2=62	60+2=62
	(Type)/(Time)				